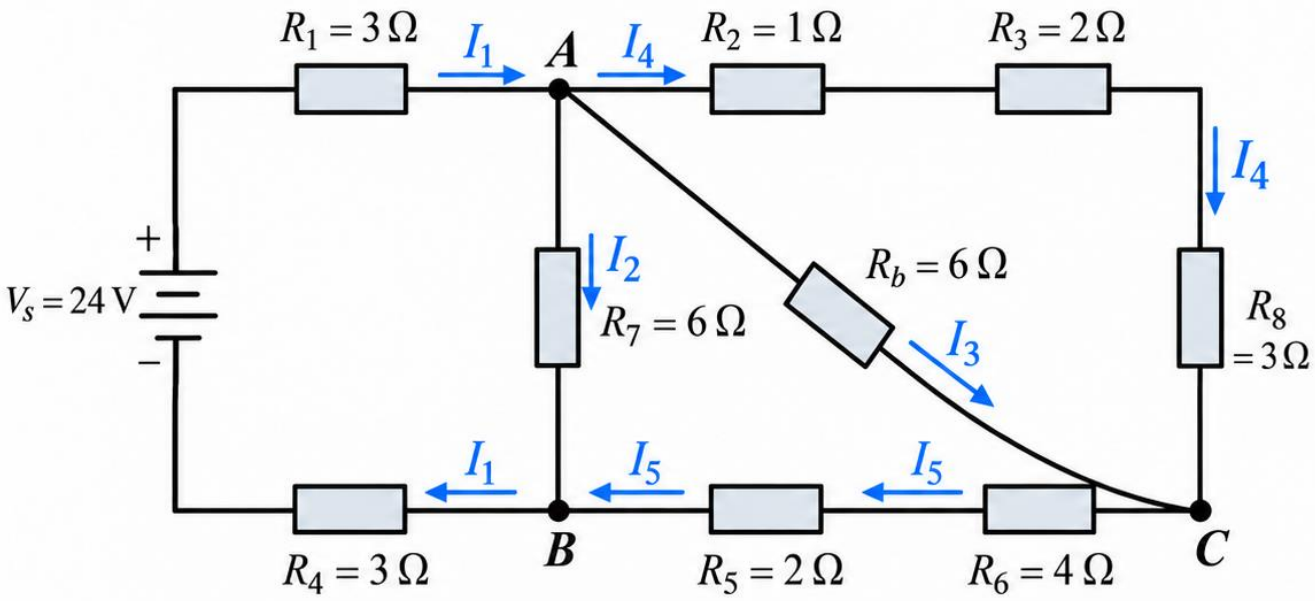
Follow the five branch currents



I_1: source, R_1 and R_4

I_2: R_7, from A to B

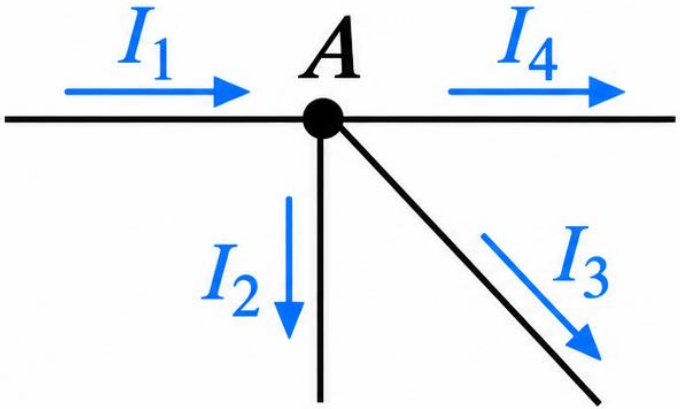
I_3: R_b, from A to C

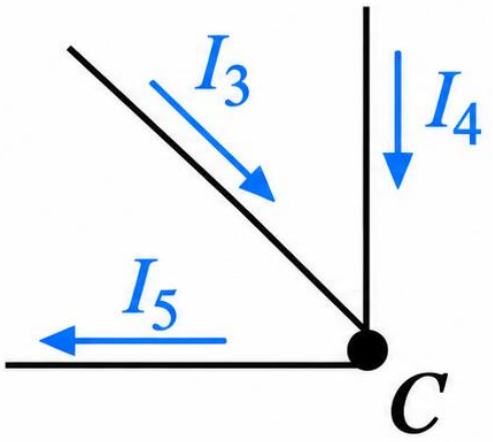
I_4: R_2, R_3 and R_8, from A to C

I_5: R_6 and R_5, from C to B

Components in one series path carry the same current.

Apply current conservation at the junctions



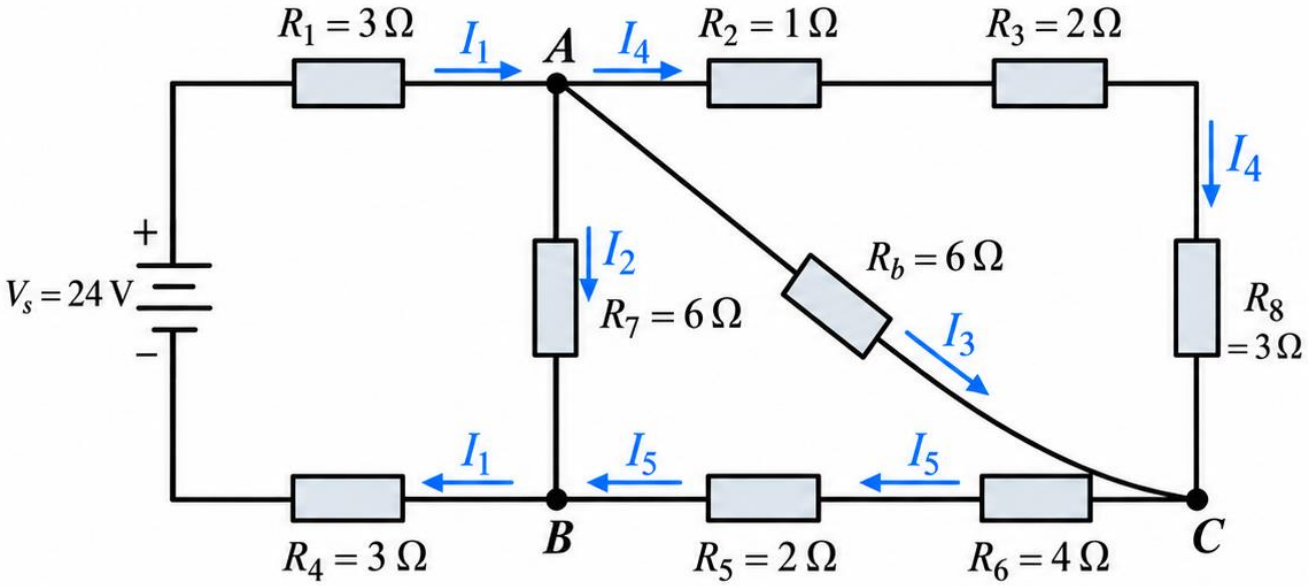


At A: $I_1 = I_2 + I_3 + I_4$ **At C:** $I_5 = I_3 + I_4$

At B: $I_2 + I_5 = I_1$ repeats the same information.

Solve I_2, I_3 and I_4 first; recover I_1 and I_5 afterwards.

Apply voltage conservation around closed paths



Left path

$$24 = (3+3)I_1 + 6I_2$$
$$24 = 6I_1 + 6I_2$$

Lower triangle

$$6I_2 = (2+4)I_5 + 6I_3$$
$$6I_2 = 6I_5 + 6I_3$$

Upper triangle

$$6I_3 = (1+2+3)I_4$$
$$6I_3 = 6I_4$$

Resistances in Ω; currents in A; every term is a voltage in V.

A closed path defines the voltage balance; the arrows define branch-current directions.

Solve three independent branch currents.

Form the reduced equations

Substitute $I_1 = I_2 + I_3 + I_4$ and $I_5 = I_3 + I_4$.

$$\begin{aligned} 12I_2 + 6I_3 + 6I_4 &= 24 \\ 6I_2 - 12I_3 - 6I_4 &= 0 \\ 6I_3 - 6I_4 &= 0 \end{aligned}$$

Unknown order

$$x = \begin{bmatrix} I_2 \\ I_3 \\ I_4 \end{bmatrix}$$

MATLAB code

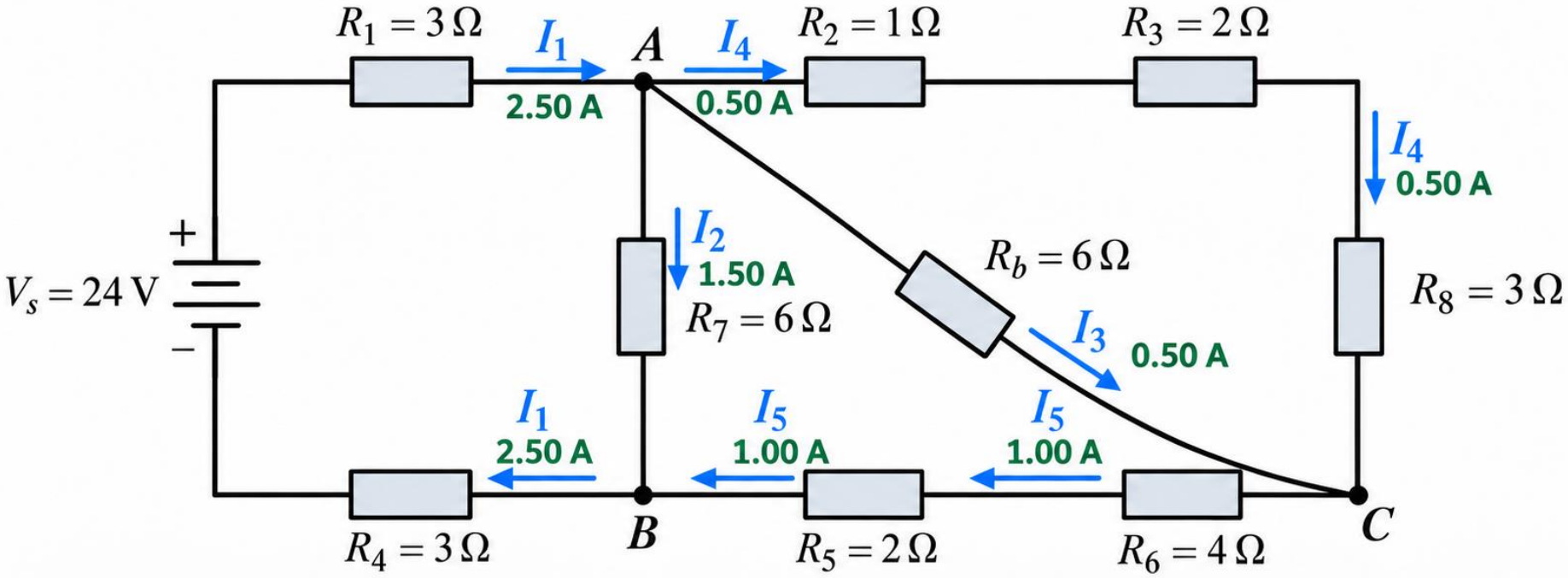
```
A = [12 6 6; 6 -12 -6; 0 6 -6];
b = [24; 0; 0];
x = A\b;
```

Result

$$x = \begin{bmatrix} 1.50 \\ 0.50 \\ 0.50 \end{bmatrix} \text{A} \qquad \begin{aligned} I_2 &= 1.50 \text{ A} \\ I_3 &= 0.50 \text{ A} \\ I_4 &= 0.50 \text{ A} \end{aligned}$$

A coefficients in Ω; b in V; x in A.

Recover and interpret all five currents



$$I_1 = I_2 + I_3 + I_4 = 2.50\ \text{A}$$

$$I_5 = I_3 + I_4 = 1.00\ \text{A}$$

All currents are positive, so conventional current follows the assumed arrows.

Check current and voltage balances.

Current balance

At A ✓

$$2.50\ \text{A} = 1.50\ \text{A} + 0.50\ \text{A} + 0.50\ \text{A}$$

At C ✓

$$1.00\ \text{A} = 0.50\ \text{A} + 0.50\ \text{A}$$

Voltage balance

Left path ✓

$$24\ \text{V} = 15\ \text{V} + 9\ \text{V}$$

Source voltage equals the total drop.

Lower triangle ✓

$$9\ \text{V} = 6\ \text{V} + 3\ \text{V}$$

Equal potential difference along either route.

Upper triangle ✓

$$3\ \text{V} = 3\ \text{V}$$

Equal potential difference along either route.

Substitute the currents into the original circuit equations.